

Terminology Silos and the Consilience Gap: A Quantitative Audit of Cross-Domain Vocabulary

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Abstract

Interdisciplinary consilience is claimed everywhere and measured almost nowhere. This paper measures one of its basic preconditions: whether scientific domains share vocabulary. We generalize the keyword-taxonomy audit of the QNFO research program (a seven-domain organization whose own canonical taxonomy proved strictly partitional — 334 of 335 keywords occur in exactly one domain) into a quantitative instrument and apply it to an external sample of six arXiv disciplines (240 recent abstracts; 466 technical compound terms). The result: terminology silos are the norm, not an organizational accident. 97.2% of the external technical vocabulary occurs in exactly one discipline; the shared core (terms in three or more disciplines) is empty; and the 2.8% of terms that do bridge disciplines are massively enriched in method-level vocabulary (machine learning, language models, upper bounds) rather than structural concepts (Fisher exact $p = 8.5e-7$, odds ratio 70). On the QNFO corpus, semantically linked papers within a program share title vocabulary (mean Jaccard 0.11 vs 0.03 random), while the cross-domain semantic bridges the program itself cites carry zero lexical signal unless an author deliberately wrote the bridge into a title. We conclude that the consilience gap is in part a vocabulary gap: shared terminology is rare, structurally biased toward methods, and built rather than emergent. The paper closes with the infrastructure response — bridge vocabularies as first-class research infrastructure, taxonomy engineering, and semantic mapping — and states where its premises end.

1. Introduction

A researcher in quantum foundations, a researcher in algebraic number theory, and a researcher in machine learning can work on the same structural object — a nested, hierarchical partition of a state space — and never discover each other. The first calls it a measurement hierarchy, the second calls it a valuation, the third calls it a cluster tree. Their vocabularies do not overlap; their results do.

This is the terminology silo problem. It is a commonplace complaint of interdisciplinary research that “we all mean the same thing but use different words.” It is rarely quantified. This paper quantifies it, at two scales: the internal scale of a single research organization (the QNFO program, seven research domains sharing one canonical keyword taxonomy) and the external scale of six scientific disciplines sampled from arXiv.

The motivation is not lexicographic. Large-scale scientific progress is increasingly bottlenecked at the vocabulary layer: literature search, citation graphs, knowledge graphs, and AI-assisted synthesis all operate primarily on lexical matching, and lexical matching systematically misses cross-domain correspondences. If the shared vocabulary of science is 3% of its technical terms, then keyword-based discovery is a 3%-recall instrument for the connections that interdisciplinary consilience depends on. That is the gap this paper measures and then proposes infrastructure for.

The stake for a reader is direct: if you search for related work and miss the field that already solved your problem under a different name, you are paying the terminology silo tax. This paper measures the size of the tax (97% of technical vocabulary is domain-local), its structure (what little is shared is method, not substance), and what to do about it (build bridge vocabularies; do not wait for them to emerge).

2. The seed audit: a seven-domain organization's own vocabulary

The QNFO research program spans seven domains — ultrametric physics (UMP), the laws of form (SLB), infomatics (INM), paradigm engineering (CFE), consilience research (RES), a cloud-native platform (PLT), and interactive demos (DEM) — and maintains a canonical keyword taxonomy of 335 terms for GitHub discovery (docs/QNFO-KEYWORD-TAXONOMY.md v1.0, 2026-08-05). The program's central claim is that the seven domains are seven vocabularies for one structural object: nested hierarchical partition logic.

A computational audit of that taxonomy (10.5281/zenodo.22071421) found the claim lexically unsupported. 334 of 335 keywords occur in exactly one program; exactly one keyword (complexity-measure) occurs in two; none occurs in three or more. The taxonomy's own bridge subsections — named bridge concepts inside each program — are program-local anchors: they name connections without instantiating shared vocabulary. A Fisher exact test of whether the four bridge families (valuation, hierarchy, distinction, bound) coincide with load-bearing vocabulary (keywords shared by three or more programs) gives $p = 1.0$: no enrichment whatsoever. The consilience, if it exists, is semantic — carried by corpus-level links between records — not lexical.

This single-case result invites the general question: is the QNFO taxonomy an organizational artifact, or a microcosm of science? The rest of this paper treats it as a hypothesis source and builds an instrument that can be applied to any domain partition.

3. Definitions and instrument

Three quantities, all computed from a domain-keyword partition (a set of domains, each with a set of technical terms):

1. **Partitionality index** — the fraction of distinct keywords occurring in exactly one domain. A partitionality of 1.0 means total vocabulary isolation; 0.0 means every term is shared.

2. **Bridge share** — the fraction of distinct keywords occurring in two or more domains. The complement of local vocabulary.
3. **Shared core** — keywords occurring in three or more domains (a load-bearing shared vocabulary in the sense of the seed audit).

Plus a structural probe: a one-sided Fisher exact test on the bridge x general-family contingency, where the general family is a curated list of structurally general method and pattern terms (machine learning, neural networks, Monte Carlo, phase transitions, upper bounds, and similar vocabulary that plausibly travels across fields). The test asks whether bridge terms are drawn preferentially from the general family rather than uniformly from all vocabulary.

Vocabulary extraction. For the external sample, technical vocabulary is extracted from paper titles and abstracts as compound terms (bigrams of content words), after removing a large stopword set (function words and generic science shells such as “model”, “system”, “theory”, “study”, “field”, “structure”). Bigrams are used because modern technical vocabulary is overwhelmingly compound (phase-transition, neural-network, monte-carlo, p-adic); single tokens are too ambiguous to discriminate domains. Per domain, the 80 most frequent bigrams (frequency at least 2 in the sample) form the domain vocabulary. The extraction is deterministic: no randomness, fixed seed policy for any sampling.

The instrument reproduces the published seed audit exactly (validation 9/9: 335 distinct keywords, 334 local, one shared-by-two, zero core, 53/282 family-bridge contingency, Fisher $p = 1.0$), so the external numbers below rest on a validated pipeline.

4. Results I: six disciplines, one vocabulary wall

Six arXiv categories were sampled (40 most recent abstracts each, fetched 2026-08-24): quantum physics (quant-ph), number theory (math.NT), machine learning (cs.LG), genomics (q-bio.GN), materials science (cond-mat.mtrl-sci), and economic theory (econ.TH). Vocabulary extraction yielded 466 distinct technical bigrams across the six domains.

Quantity	Value
Distinct technical terms	466
Domain-local terms (exactly one domain)	453
Partitionality index	0.9721
Bridge terms (two or more domains)	13
Bridge share	0.0279
Shared core (three or more domains)	0

The partitionality hypothesis (H-SILO-1: at least 90% of vocabulary domain-local) is confirmed: 97.2%. The shared core is empty at the three-domain threshold. Interdisciplinary rhetoric notwithstanding, six disciplines share essentially no technical vocabulary beyond chance.

The 13 bridge terms are the interesting residue:

- **Method vocabulary:** machine-learning, language-model, language-models, foundation-models, upper-bound (5 of 13).
- **Shared objects:** boron-nitride, quantum-defects, defects-zno, double-substitutional, optically-quantum, candidate-quantum, candidates-optically (7 of 13 — two disciplines working on the same materials).
- **Shared application:** drug-discovery (1 of 13).

The structural probe quantifies the method bias: of the 13 bridges, 5 are method/pattern terms; of the 453 domain-local terms, only 4 are. The Fisher exact test on the bridge x general-family contingency gives $p = 8.5e-7$ with odds ratio 70.2 — bridge vocabulary is seventy times more likely to be method vocabulary than domain-local vocabulary is. What travels between disciplines is not the structural substance (hierarchies, valuations, measures) but the toolkit (learning, models, bounds).

The QNFO seed case shows the same structure in extremis: bridge share 0.0030, and its own bridge families show zero enrichment against load-bearing vocabulary ($p = 1.0$). The seed's bridge vocabulary is domain-anchored (ostrowski-theorem, idele-class-group) rather than method-level — an organization whose bridge infrastructure is homegrown, not imported.

5. Results II: the QNFO corpus — semantic links and lexical silence

The external sample measures vocabulary; the QNFO corpus (578 titled records in the living paper) measures whether semantic relatedness is lexically visible. Three probes.

Probe 1 — the knowledge-graph link network is program-local. The QNFO knowledge graph's paper-to-paper semantic edges (CITES, DEPENDS_ON, MOTIVATES, REFINES, BRIDGES, REFERENCES, SUPERSEDES, LINKS_TO, RELATES_TO) resolve to 40 pairs of corpus records. Classifying each record into its program by title-vocabulary overlap with the taxonomy, 8 pairs are same-program and 0 pairs are cross-program (32 pairs involve at least one unclassifiable endpoint). The graph links papers within programs; it has not yet built the cross-program network. The link infrastructure is itself siloed.

Probe 2 — within-program links are lexically visible. Same-program linked pairs share title vocabulary at mean Jaccard 0.115, against 0.028 for random pairs (seeded, same cardinality): a +0.086 elevation. Within a program, shared vocabulary tracks semantic relatedness. This is the partitional null: vocabulary signals structure only inside a domain.

Probe 3 — the cited cross-domain bridges carry zero lexical signal. The umbrella program paper (10.5281/zenodo.22073477, section 5.1) cites three corpus records as the semantic bridges that carry the consilience: measurement stratigraphy linking epistemology to valuation theory (10.5281/zenodo.21705220), the valuation-without-reals framework (10.5281/zenodo.21803677), and ultrametric topology in semantic memory (10.5281/zenodo.19564091). The first pair (measurement-stratigraphy vs adelic-

shannon-theory) has title-vocabulary Jaccard **0.0000**; the third pair (silent-radix cryptography vs ultrametric numeration) also 0.0000. The consilience is lexically invisible where no author built the bridge.

The single visible exception proves the rule: the valuation-without-reals record shares the token “valuation” with measurement stratigraphy at Jaccard 0.333 — because its author deliberately wrote the bridge into the title (“Valuation Without R”). Vocabulary bridges are built, not emergent.

Summary of hypothesis outcomes.

Hypothesis	Prediction	Result
H-SILO-1	partitionality ≥ 0.90	Confirmed (0.9721 external; 0.9970 seed)
H-SILO-2	bridge share < 0.10 ; Fisher $p < 0.05$	Confirmed (0.0279; $p = 8.5e-7$, OR 70.2)
H-SILO-3	semantic links carry no lexical signature	Supported (0.0 on non-authored bridges; within-program links visible)

6. Why silos persist

The measurement explains the phenomenon; the persistence mechanisms are incentive and infrastructure. Four evidence-grounded mechanisms:

1. **Field-local prestige economies.** Publication, hiring, and funding reward vocabulary mastery within a field; mastery is signaled by using the field’s terms, not by translating them. The QNFO seed shows the organizational version: programs are rewarded for domain depth, and the taxonomy’s bridge subsections exist precisely because nothing in the ordinary keyword flow crosses boundaries.
2. **Vocabulary gatekeeping by venues.** Journals, conferences, and archives curate keyword taxonomies that are field-scoped. A term that does not appear in the venue’s taxonomy is invisible to its search; authors optimize for the venue’s vocabulary, deepening the partition.
3. **Translation costs.** Establishing that “valuation” in number theory corresponds to “measurement hierarchy” in epistemology costs verification effort that the author bears alone and the field does not reward. The QNFO corpus demonstrates the result: where an author paid the cost (the “Valuation Without R” title), the bridge is visible; elsewhere it is not.
4. **Path dependence in taxonomy formation.** Keyword taxonomies grow by accretion inside domains; they are rarely audited for cross-domain overlap, and no one owns the intersection. The seed audit is itself the exception that tests the rule: it took a computational audit to discover that an organization’s own vocabulary was 99.7% partitional.

7. What the silo costs

The measurable proxies from this study bound the cost:

- **Discovery failure.** With a 2.8% bridge share, keyword search for a concept outside the searcher's home domain has a recall ceiling of a few percent for cross-domain correspondences. The exemplar result (Jaccard 0.0 between two records the program itself cites as semantically linked) shows the failure at record level: the linkage exists in prose, not in vocabulary, so it is invisible to lexical retrieval.
- **Duplicated discovery.** Fields that cannot see each other's vocabulary cannot see each other's results; the same structural result is re-derived under new names. The method-bias finding (bridges are 70x enriched in method vocabulary) sharpens this: disciplines import each other's tools while remaining blind to each other's structures.
- **Convergence delay.** Cross-domain consilience — the recognition that two fields describe one object — is the rate-limiting step for large-scale synthesis. If vocabulary is 97% partitional, consilience is never discoverable bottom-up; it requires an explicit act of bridge-building.

8. Consilience infrastructure: bridges are built, not emergent

The constructive response follows directly from the measurement. If shared vocabulary is rare, method-biased, and author-made, then consilience needs deliberate vocabulary infrastructure:

1. **Bridge vocabularies as first-class artifacts.** A bridge vocabulary is a maintained, versioned mapping between the terms of two or more domains, with verification notes (which terms correspond, at what level of structural fidelity). The QNFO taxonomy's bridge subsections are a primitive version; the corpus's title-visible bridges are the worked examples of the practice.
2. **Taxonomy engineering with an intersection owner.** Every domain taxonomy should have a named owner of the intersection: a person or process that periodically audits the partition, measures bridge share, and curates the cross-domain terms. The instrument in this paper is the audit tool; the seed validation shows it can be run continuously at organizational scale.
3. **Semantic mapping between field vocabularies.** For the AI-assisted synthesis pipeline, the fix is a mapping layer between field-specific vocabularies — the semantic analogue of a bilingual dictionary — so that retrieval can find “measurement hierarchy” when asked for “valuation”. The H-SILO-3 result quantifies the need: without the mapping layer, cross-domain retrieval operates at zero recall on the program's own cited bridges.
4. **Title-visible bridges.** The cheapest infrastructure is authorial: when a paper connects domains, name the connection in the title. The one visible exemplar (Jaccard 0.333) shows the effect; it is the only corpus bridge that lexical retrieval can find.

9. Where the premises end

This paper’s claims are derived from the measurement pipeline; its premises end at four named inputs:

- **P1 — the vocabulary model.** Technical vocabulary is modeled as stop-word-filtered bigrams from titles and abstracts. This is a proxy for “the terms a field uses”: it excludes single-token technical terms, terms below the frequency threshold, and terms that appear only in bodies. The QNFO seed, which uses curated keywords, shows the same partitional structure, but the two pipelines are not identical instruments.
- **P2 — the domain partition.** The external sample partitions science by arXiv category. Real disciplines overlap and subdivide; the partition choice affects the measured quantities (finer partitions raise partitionality).
- **P3 — the general-family list.** The Fisher enrichment test depends on the curated method/pattern list. The list is conservative (method terms only) and published in full with the instrument; re-running with a different list changes the odds ratio, not the qualitative result (bridges are method-concentrated by inspection of all 13 terms).
- **P4 — the corpus scope.** The QNFO corpus probes are single-organization evidence: 578 records, 40 resolvable graph pairs, three cited bridges. The external generalization rests on the six-discipline vocabulary sample (240 abstracts).

The quantitative claims (partitionality, bridge share, Fisher p, Jaccard values) are reproduced by the deposited scripts with fixed seeds (artifacts/verification/). No claim here extends beyond the measured samples; the mechanisms of Section 6 are explanatory hypotheses grounded in those measurements plus standard economics of science, not independent estimates.

10. Limitations and next steps

Limitations: sample breadth (six disciplines, 40 abstracts each — a snapshot, not a census); the single-token blind spot of the bigram model; the absence of a temporal dimension (vocabulary silos may be widening or narrowing over time); and the organization-scale corpus probes.

Next steps: (1) scale the external sample to 30+ disciplines and 1,000+ abstracts per discipline, with author-keyword metadata where available; (2) add the temporal dimension via arXiv date-banded samples to measure whether bridge share is changing; (3) build and evaluate the semantic mapping layer (bilingual-dictionary analogue) on the QNFO corpus, testing whether mapped retrieval recovers the program’s own cited bridges; (4) publish the bridge-vocabulary audit as a recurring instrument for the QNFO program (the seed case already runs it).

11. Reproducibility

- `scripts/arxiv_domain_sample.py` — external evidence collection (arXiv API, deterministic extraction; raw evidence in `artifacts/external-search/arxiv_raw_2026-08-24.json`).

- `scripts/build_qnfo_domains.py` — canonical taxonomy parser (method of `rq5_keyword_load.py`).
- `scripts/terminology_silos.py` — measurement toolkit (partitionality, bridge share, shared core, Fisher exact).
- `scripts/hsilo3_semantic_links.py` — corpus probes (KG structure, same-vs-random, exemplar bridges).
- `scripts/validate_seed_vs_published.py` — 9/9 validation against the published audit.
- All outputs and logs: `artifacts/verification/`. Seeds fixed (42 where sampling occurs); the extraction itself is deterministic.

References

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